

DATASET INSPECTION REPORT

Project: Business Analytics Case Study

Dataset: Telco Customer Churn Dataset

Programme: AnalystLab Africa Data Analytics Internship

Week: 1

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Dataset Overview: The dataset used for this project is the Telco Customer Churn Dataset, which contains information about customers, the services they use, their billing details, contract information, and whether they have left the company.

The purpose of inspecting the dataset is to understand its structure, identify data quality issues, and ensure it is suitable for analysis before creating visualizations and drawing business insights.

Dataset Understanding

Item	Value
Dataset Name	Telco Customer Churn Dataset
Source	Kaggle
File Type	CSV
Number of Rows	7,043
Number of Columns	21

Missing Values / Duplicate Records: No missing values or duplicate records were found or identified after cleaning, indicating that each customer appears only once in the dataset.

Data Types: The dataset contains different types of variables as shown below.

Data Type	Examples
Text	Customer ID, Gender, Senior Citizen, Partner, Payment Method, Internet Service, Contract, Churn.
Integer	Tenure
Decimal	Monthly Charges, Total Charge

The Data types were reviewed and corrected where necessary to ensure accurate statistical analysis.

Summary Statistics: The descriptive statistics were calculated for the numerical variables. The following statistics were generated:

- Count
- Mean
- Median
- Mode
- Minimum
- Maximum

- Range
- Standard Deviation
- Skewness
- Kurtosis

These statistics provide a better understanding of how customer charges and tenure are distributed throughout the dataset.

Data Quality Observations

Several checks were carried out to assess the quality of the dataset.

Findings

- Column names are consistent and clearly defined.
- Numerical columns were correctly formatted.
- Categorical values were standardized.
- Missing values were addressed.
- Duplicate records were checked.
- Data types were corrected where necessary.
- The dataset is complete and ready for exploratory data analysis.

Conclusion

The Telco Customer Churn dataset was successfully inspected and cleaned before analysis.

The inspection confirmed that the dataset has an appropriate structure, contains relevant customer information, and is suitable for business analytics. After addressing any identified data quality issues such as missing values or incorrect data types, the dataset is considered reliable for exploratory data analysis, visualization, and the development of business insights and recommendations regarding customer churn.

The cleaned dataset provides a solid foundation for identifying customer behaviour patterns, understanding factors influencing churns, and supporting data-driven business decisions.